RAMCO INSTITUTE OF TECHNOLOGY
Rajapalayam

Department of Computer Science and Engineering
Academic Year: 2020- 2021 (Even Semester)

Year, Semester & Branch: VI Semester B.E. Computer Science and Engineering.
Course Code & Title: CS8601 Mobile Computing
Name of the Faculty member: Dr.I. Gethzi Ahila Poornima

Theme of discussion: Mobile Payment System.

Date and Time: 17.05.2021 & 10.00 am to 10.50 am

Innovative practices: JIGSAW Collaborative Learning

"Jigsaw classroom" [1] is a research based learning technique invented and developed in the early 1975. In this type of activity each student becomes a resource. Students learn through the process of communicating with one another about a given skill or procedure, topic or problem.

Topic: Mobile Payment System

Sub Topics:

- Mobile Payment System
- Security issues in Mobile Payment System
- Mobile Operating system
- M-Commerce

Outcome

- Describe the Mobile Payment System and M-Commerce

Justification for choosing the topic

This topic describes the basics about the mobile payment system and M-commerce. As a computer science engineering student, he/she should know about the mobile apps as well as the security issues, and pros and cons of mobile payment system. As a result of this activity the students get exposure to the mobile payment systems available in market. Students can master all the available mobile payment system apps, their pros and cons and especially security issues.

Procedure and Implementation

Team formation

The heterogeneous groups with six members were formed. Each group formed by their own willingness. One group contains 4-5 members. The total class strength is 49 (Boys and Girls). Twelve heterogeneous groups were formed based on willingness of the students.

Sno	Group name	No. of members
1	Jigsaw Group 1	4
2	Jigsaw Group 2	4
3	Jigsaw Group 3	4
4	Jigsaw Group 4	4

5	Jigsaw Group 5	5
6	Jigsaw Group 6	4
7	Jigsaw Group 7	4
8	Jigsaw Group 8	4
9	Jigsaw Group 9	4
10	Jigsaw Group 10	4
11	Jigsaw Group 11	4
12	Jigsaw Group 12	4

The figure 1 and figure 2 shows the group name and its team leader which is created in Canvas.

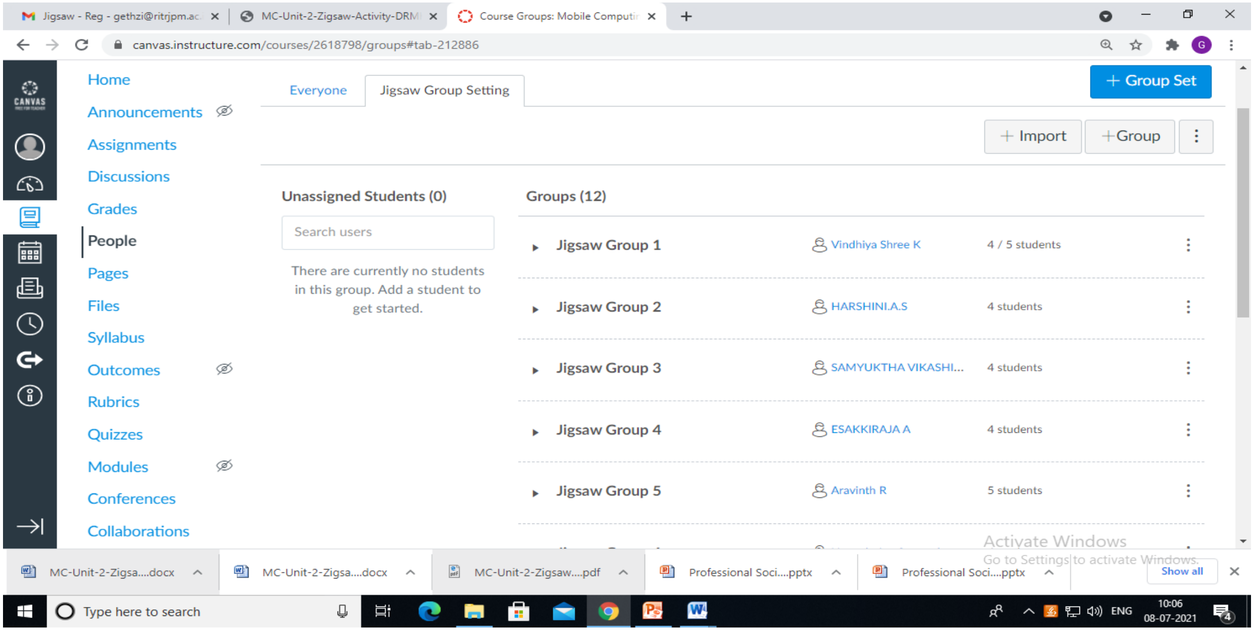


Figure 1: Group name and its team leader

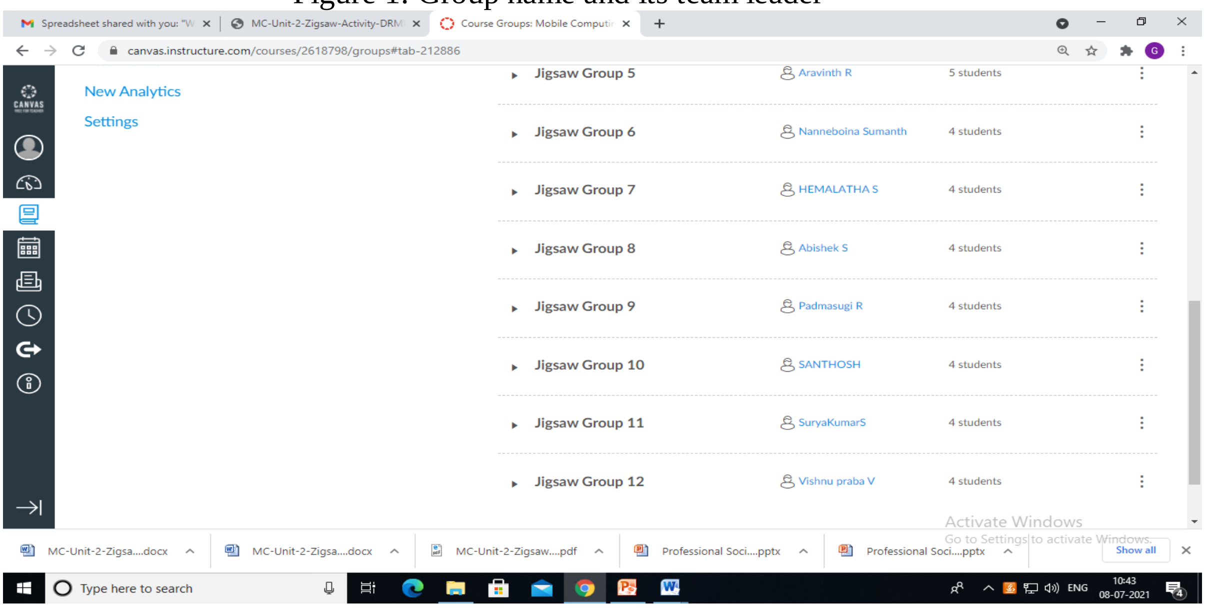


Figure 2: Group name and its team leader

Step 1

- Twelve groups with four/five members were formed (online) based on the students'academic performance and ability. It is called home group.
- One student from each group selected as the leader. Each student in a group was given a number from 1 to 4. Each number is representing the following four sub topics. A student was assigned one subtopic in a home group.
  1. Security issues in Mobile Payment System
  2. Mobile Payment System
  3. Mobile Operating system
  4. M-Commerce

Step 2

- Expert groups were formed that consist of student from home groups who were assigned the same number in a separate Google meet link.
- Expert group members were given 15 minutes to discuss an assigned sub topic.

Step 3

- The expert group students brought back into their home groups and allotted 10 minutes for further discussion in which each member become expert in the topic said above.

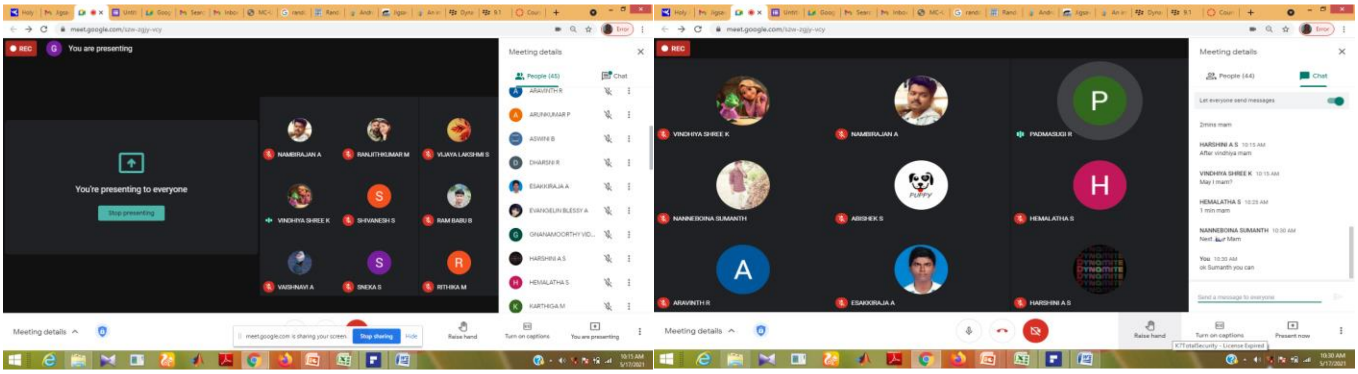
Step 4

- 10 minutes were allotted for the presentation. One student from each home group presented his learned content.
- The others in the group were encouraged to raise queries for clarification

Step 5

- At the end of the session, feedback was collected by sharing online feedback form. The analysis report is given in Figure 4.

Glimpse



Reflective Report

Challenge

- Participation of slow learner for discussion on the sub-topic.

Observations:

The students were provided the opportunity to become experts in particular sub topic. They shared the knowledge they gained with peers. This activity promotes both self- learning and peers learning among the students to master the course content at a deeper level. Most of the students participated actively. The Jigsaw technique helped the students to develop expertise in a particular topic.

Students from all the groups expressed their knowledge and enhanced their peer learning and teaching.

Students Response:

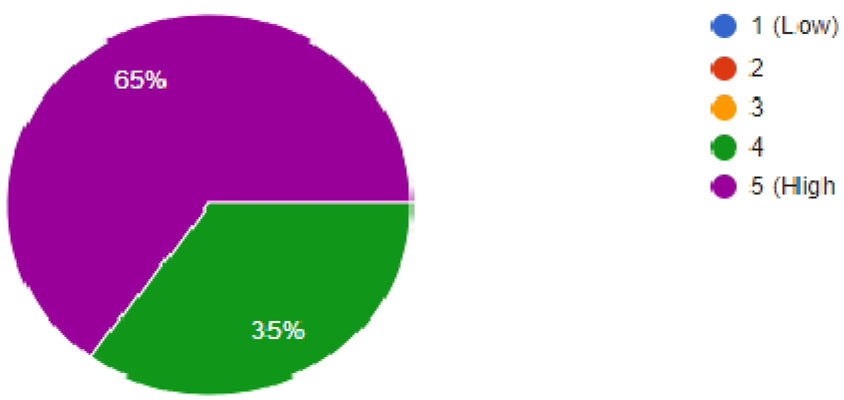
- Most of the students discussed with their teammates comfortably during the discussion inexpert groups.
- All the students actively involved with peers to discuss the topics in their home group.
- Most of the students enjoyed the discussion in the home and expert group.
- 85% of the students gave excellent feedback regarding the feedback session. Figure 4 shows the feedback analysis report.

Relevance of program outcomes

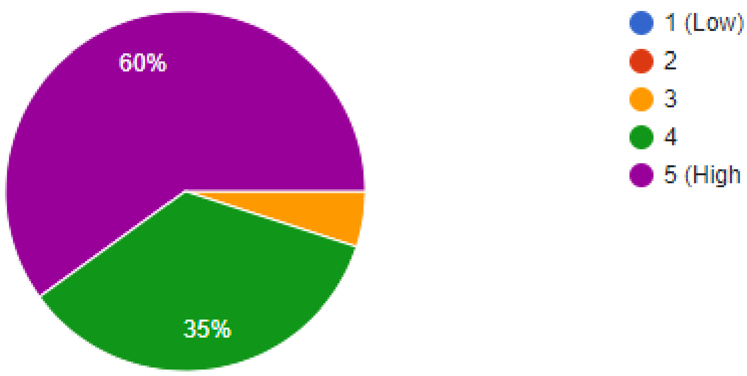
Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Outcome	3						3	2	2	3		2

Figure 4: Sample Feedback Analysis

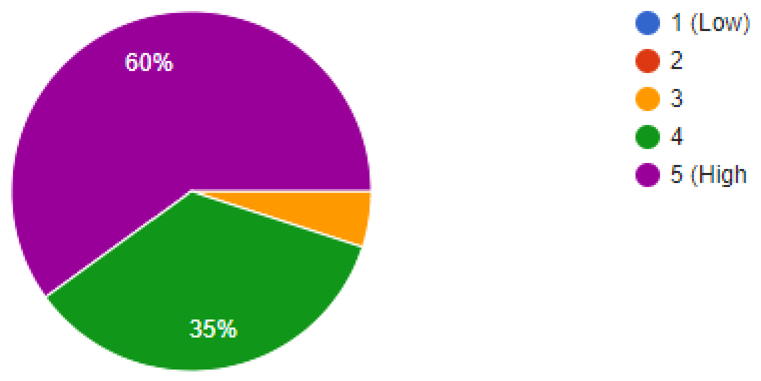
Q1. Rate the Jigsaw Activity



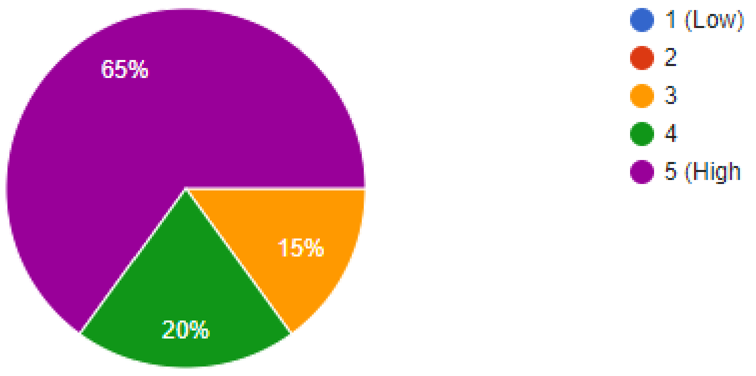
Q2. Comprehension Building



Q3. Improvement in Learning, Communication



Q4. Enjoyment in Learning



Reference:

1. https://www.newcastle.edu.au/_data/assets/pdf_file/0016/109600/Jigsaw-learning-activity.pdf
2. https://www.jigsaw.org/
3. Dr.M.Kaliappan, CS8601 – Mobile Computing Topic: GSM Services and architecture, Jigsaw activity, 10.02.2020

Faculty In charge

Dr. I. Gethzi Ahila Poornima

HOD/CSE

Dr.K.Vijayalakshmi